Gravitation $w_{1+\infty}$ Symmetry

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Abstract

Extremely rough notes on gravitation $w_{1+\infty}$ symmetry. Maybe these will turn into something coherent, maybe not!

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1 BMS Invariance of Gravitational Scattering

1.1 The Phase Space of Interest

Any 3+1-dimensional spacetime can be locally put into Bondi gauge

$$ds^2 = -Udu^2 - 2e^{2\beta}dudr + g_{AB}(dx^A - U^A du)(dx^B - U^B du) \tag{1}$$

subject to the areal distance condition

$$\partial_r \det\left(\frac{g_{AB}}{r^2}\right) = 0. \tag{2}$$

An asymptotically flat spacetime can, near future null infinity, be more strongly put into coordinates such that

$$\begin{aligned} ds^2 = &-du^2 - 2dudr + 2r^2\gamma_{z\bar{z}}dzd\bar{z} \\ &+ \frac{2m_B}{r}du^2 + rC_{zz}dz^2 + rC_{\bar{z}\bar{z}}d\bar{z}^2 - 2U_z dudz - 2U_{\bar{z}}dud\bar{z} + \cdots . \end{aligned} \tag{3}$$

If we rewrite the tranverse / angular coordinates $(z, \bar{z})$ as (x^A), we have equivalently

$$\begin{aligned} ds^2 = &\left(-1 + \frac{2m_B(u,z,\bar{z})}{r} + \mathcal{O}\left(\frac{1}{r^2}\right)\right)du^2 - 2\left(1 + \mathcal{O}\left(\frac{1}{r^2}\right)\right)dudr \\ &+ \left(r^2 q_{AB}(u,x) + rC_{AB}(u,x) + \mathcal{O}(1)\right)dx^A dx^B \\ &- 2\left(U_z(u,z,\bar{z}) + \mathcal{O}\left(\frac{1}{r}\right)\right)dudz + \text{c.c.} \end{aligned} \tag{4}$$

Here q_{AB} is the leading-order transverse metric

$$2r^2\gamma_{z\bar{z}}dzd\bar{z} = q_{AB}dx^Adx^B \tag{5}$$

and the areal distance condition forces the shear C_{AB} to be traceless w.r.t. this metric[1]

$$q_{AB}C^{AB} = 0 \tag{6}$$

which is exactly why we have no $C_{z\bar{z}}$ terms. Observe that this is very nearly the statement that the components of the metric are the flat-space Bondi components plus the most general polynomial decay; the only strengthenings are the traceless shear condition coming from the areal distance gauge choice, plus the slightly strong fall-off on g_{ur} which turns out to come from insting on finite energy flux[2] through future null infinity $\mathcal{I}^+$. In fact we will need to make further restrictions, namely

$$U_z = -\frac{1}{2}D^zC_{zz} \tag{7}$$

and the conjugated equation. These come from insisting that the Weyl tensor decay sufficiently quickly near $\mathcal{I}^+$, which is necessary for a conformal compactification. In fact this condition give a tower of additional constraints at further subleading order in r, but we won't need to examine those for this work.

The above discussion has a few subtle quantifiers, so we phrase our phase space of interest carefully. First, we assume we have some reasonable formal definition of future null infinity $\mathcal{I}^+$, which should roughly be "where null rays eventually end up, if they don't hit a singularity or get trapped first." We assume that $\mathcal{I}^+ \simeq S^2 \times \mathbb{R}$, as is the case for flat Minkowski. A spacetime is then asymptotically flat if there exist Bondi-gauge coordinates $(u, r, z, \bar{z})$ near $\mathcal{I}^+$ such that the the metric falls off as described above[3], plus a corresponding set of coordinates near past null infinity $\mathcal{I}^-$ with the analogous properties. Our phase space is exactly the set of asymptotically flat spacetimes in 3+1 dimensions.

1.2 Supertranslations

We are interested in diffeomorphisms which preserve the form of the metric (3). As a side note, it is nearly universal in physics, when discussing a map in coordinates of the form $x \mapsto x'$, to fail to distinguish between a mere change of coordinates and a diffeomorphism. Even more unfortunately, it is common especially in classical literature to always speak of coordinate changes. This is wrong! A coordinate change is always a trivial thing to do, while a diffeomorphism moves points around in spacetime. Here we truly are goin to perform diffeomorphisms while maintaining a fixed coordinate system, so that $x'(x)$ means "the coordinates of the image of the point which had coordinates x. This distinguish doesn't matter if we just want to shut up and calculate, but we do not.

Now as usual in physics, we simplify by studying infinitesimal diffeomorphisms, or equivalently pushforwards under vector fields $\xi = \xi^a\partial_a$ w.r.t. whose components we linearize. For now we want ξ to have at most $\mathcal{O}(1)$ norm near $\mathcal{I}^+$ (we want these fields to generalize translations; soon we will relax this condition to get a generalization of rotations and boosts). So we will require

$$\xi^u, \xi^r \in \mathcal{O}(1), \quad \xi^z, \xi^{\bar{z}} \in \mathcal{O}\left(\frac{1}{r}\right) \tag{8}$$

or more explicitly

$$\begin{aligned}
\xi^u(u,r,z,\bar{z}) &= \sum_{n=0}^{\infty}\frac{\xi^{r(n)}(u,z,\bar{z})}{r^n}, \quad \xi^r(u,r,z,\bar{z}) = \sum_{n=0}^{\infty}\frac{\xi^{u(n)}(u,z,\bar{z})}{r^n}, \\
\xi^z(u,r,z,\bar{z}) &= \sum_{n=1}^{\infty}\frac{\xi^{z(n)}(u,z,\bar{z})}{r^n}, \quad \xi^{\bar{z}}(u,r,z,\bar{z}) = \sum_{n=1}^{\infty}\frac{\xi^{\bar{z}(n)}(u,z,\bar{z})}{r^n}
\end{aligned} \tag{9}$$

[1] Plus an infinite tower of additional conditions at subleading order in r

[2] Is this the right observable?

[3] Plus further constraints at subleading orders which, again, we won't need here.

In general, the variation of the metric (or any other same-rank tensor) under an infinitesimal diffeomorphism V is given by

$$\delta_V g_{ab} = \mathcal{L}_V g_{ab} = V^c \partial_c g_{ab} + g_{ac}\partial_b V^c + g_{bc}\partial_a V^c. \tag{10}$$

For a metric of the form considered here, we have for example

$$\begin{aligned}
\delta_\xi g_{uu} &= \xi^c \partial_c g_{uu} + 2 g_{uc}\partial_u \xi^c \\
&= \left(\xi^r \partial_r + \xi^u \partial_u + \xi^z \partial_z + \xi^{\bar{z}} \partial_{\bar{z}}\right) g_{uu} \\
&\quad + 2 g_{uu}\partial_u \xi^u + 2 g_{ur}\partial_u \xi^r + 2 g_{uz}\partial_u \xi^z + 2 g_{u\bar{z}}\partial_u \xi^{\bar{z}} \\
&= 2 g_{uu}\partial_u \xi^u + 2 g_{ur}\partial_u \xi^r + \mathcal{O}\left(\frac{1}{r}\right) \\
&= -2\partial_u \xi^u + -2\partial_u \xi^r + \mathcal{O}\left(\frac{1}{r}\right).
\end{aligned} \tag{11}$$

Here we have assumed the pre-variation metric g_{ab} has the form considered above. Similarly,

$$\begin{aligned}
\delta_\xi g_{uz} &= \xi^c \partial_c g_{uz} + g_{uc}\partial_z \xi^c + g_{zc}\partial_u \xi^c \\
&= \xi^u \partial_u g_{uz} + g_{uu}\partial_z \xi^u + g_{ur}\partial_z \xi^r + g_{zu}\partial_u \xi^u + \mathcal{O}\left(\frac{1}{r}\right) \\
&= -\xi^u \partial_u U_z - \partial_z \xi^u - \partial_z \xi^r - U_z \partial_u \xi^u + \mathcal{O}\left(\frac{1}{r}\right)
\end{aligned} \tag{12}$$

Further similar calculations give

$$\begin{aligned}
\mathcal{L}_\xi g_{ur} &= -\partial_u \xi^u + \mathcal{O}\left(\frac{1}{r}\right), \\
\mathcal{L}_\xi g_{zr} &= r^2 \gamma_{z\bar{z}} \partial_r \xi^{\bar{z}} - \partial_z \xi^u + \mathcal{O}\left(\frac{1}{r}\right), \\
\mathcal{L}_\xi g_{z\bar{z}} &= r\gamma_{z\bar{z}}\left[2\xi^r + r D_z \xi^z + r D_{\bar{z}} \xi^{\bar{z}}\right] + \mathcal{O}(1).
\end{aligned} \tag{13}$$

Less interestingly,

$$\delta_\xi g_{rr} = \mathcal{O}\left(\frac{1}{r}\right). \tag{14}$$

We want to determine which vector fields ξ respect the fall-offs and the Bondi gauge conditions. For example, maintaining the condition

$$g_{ur} = -1 + \mathcal{O}\left(\frac{1}{r}\right) \tag{15}$$

forces

$$-\partial_u \xi^u = \mathcal{O}\left(\frac{1}{r}\right) \tag{16}$$

hence

$$\xi^u(u, r, z, \bar{z}) = f(z, \bar{z}) + \mathcal{O}\left(\frac{1}{r}\right) \tag{17}$$

for some function f on S^2. Applying the result of the conditions (and carefully tracking Christofells in manipulating the sphere-covariant derivatives $D_z, D_{\bar{z}}$) gives us the general solution

$$\xi = \xi^a \partial_a = f\xi_u - \frac{1}{r}\left(D^z f \partial_z + D^{\bar{z}} f \partial_{\bar{z}}\right) + D^z D_z f \partial_r + (\text{componentwise-subleading terms}) \tag{18}$$

We can convince ourselves that this moves us between genuinely distinguishable physical states using the language of scattering. For example, suppose we have a scattering process with two outgoing massless states (photons, for simplicity) which hit $\mathcal{I}^+$ simultaneously at $u = 0$ but at different angles $(z_i, \bar{z}_i)_{i=1,2}$. Then supertranslating

everything around under any f with $f(z_1, \bar{z}_1) \neq f(z_2, \bar{z}_2)$ gives a new history where the null states exit non-simultaneously.

Under such a variation, the Bondi data changes as

$$\begin{aligned} \delta_f N_{zz} &= f\partial_u N_{zz}, \\ \delta_f m_B &= f\partial_u m_b + \frac{1}{4}\left[N^{zz}D_z^2 f + 2D_z N^{zz} D_z f + \text{c.c.}\right], \\ \delta_f C_{zz} &= f\partial_u C_{zz} - 2D_z^2 f \end{aligned} \tag{19}$$

where we have also induced the Bondi news

$$N_{AB} \equiv \partial_u C_{AB} \tag{20}$$

which encodes gravitational waves.

To gain some more intuition for how these work, we suppose our initial spacetime is flat Minkowski space, with $m_b = N_{zz} = C_{zz} = 0$. We have not yet performed a canonical analysis, but in whatever formalization you like this should be thought of as a vacuum state of (semi)classical gravity. In this case we have

$$\begin{aligned} \delta_f N_{zz} &= 0, \\ \delta_f m_B &= 0, \\ \delta_f C_{zz} &= 2D_z^2 f. \end{aligned} \tag{21}$$

In fact, it turns out that vanishing of Riemann curvature forces

$$C_{zz} = -2D_z^2 C \tag{22}$$

for some scalar field C on $\mathcal{I}^+$, so we have more neatly

$$\delta C = f. \tag{23}$$

There are nice words we can put around this: supertranslation is a physical symmetry taking us from one vacuum to another physically distinguishable one, and C is the Goldstone boson associated with this degeneracy. Observe that C is only defined modulo shifts

$$C \simeq C + \text{Ker}(D_z^2) \tag{24}$$

But this kernel contains the $\ell = 0, 1$ modes on S^2, which are exactly the f's corresponding to rigid bulk translation. So we see that bulk rigid translations does not shift C. Should we think of these, too, as large gauge transformations? Or are they only large for $\ell \geq 2$?

1.3 Abstract Comments

Now some more abstract comments. First, any vector field ξ of the form just discussed may be call a supertranslation, but really we are interested in the bona fide physical symmetries that take us between physically distinguishable states. By this criterion, any two such vector fields ξ, ξ' with the same f but which disagree at lower order in r are the same up to a small gauge transformation; typically when we refer to a "supertranslation" we really mean such an equivalence class, or equivalently the induced physical symmetry. The supertranslations (or really, the finite transformations induced by exponentiating the vector fields ξ) form an abelian group isomorphic to $C^\infty(S^2)$ under addition.

$$(\text{supertranslations}) \simeq C^\infty(S^2). \tag{25}$$

If we drop the requirement that $|\xi|^2 \in \mathcal{O}(1)$, we get a larger group that now includes *superrotations*, and we call the whole thing BMS_+. Recall that the Poincaré group has the semidirect product structure

$$(\text{Poincaré}) \simeq SO(3,1) \rtimes \mathbb{R}^4 \tag{26}$$

where by $\mathbb{R}^4$ we mean the additive group of rigid translations. Given an abstract group isomorphic to Poincaré, we can uniquely identify the $\mathbb{R}^4$ as the maximal abelian subgroup; this is taken to itself under conjugation by a Lorentz transformation, so there is no ambiguity (physically, any two observers will agree on which transformations are translations). On the other hand, there are many non-canonically-isomorphic copies of $SO(3,1)$, related by conjugation. Equivalently, two observers will agree on which transformations are in $SO(3,1)$ iff they agree on the spacetime origin p, as their copy is exactly the group $SO(3,1)_p$ of Poincaré transformations fixing p.

Something similar happens here. BMS_+ transformations (that is, BMS transformations performed near $\mathcal{I}^+$) have the same semidirect product structure, except that the abelian group of translations has been enlarged to include all supertranslations.

$$\mathrm{BMS}_+ = (\text{Lorentz}) \rtimes (\text{supertranslations}) \simeq SO(3,1) \rtimes C^\infty(S^2). \tag{27}$$

As before, we can uniquely identify the supertranslation subgroup, but there are many distinct subgroups isomorphic to $SO(3,1)$ and no observer-independent way of picking out a canonical one.

1.4 Cauchy Data

At leading order, our solution is characterized by C_{zz}, N_{zz}, m_B and their complex conjugates, all of which are maps from $\mathcal{I}^+ \simeq S^2 \times \mathbb{R}$. These are not all independent, of course. For one, we have by definition

$$N_{zz}(u,z,\bar{z}) = \partial_u C_{zz}(u,z,\bar{z}). \tag{28}$$

It turns out that N_{zz} can be any function, constrained at most to decay fast enough at $u \to \pm\infty$ if we want our solution to corresponding to a scattering process with finite-energy incoming and outgoing states. In fact, the two real degrees of freedom living in the symmetric + traceless tensor N_{AB} correspond exactly to the two polarizations of outgoing gravitational radiation. So we take the function $N_{zz}(u,z,\bar{z})$ as our first independent piece of Cauchy data.

Given N_{zz}, C_{zz} is determined up to an integration constant which we freely take to live at $\mathcal{I}^+_-$, the asymptotic past of $\mathcal{I}^-$. We have already argued that vanishing curvature forces C_{zz} to be exact in the sense that

$$C_{zz} = -2D_z^2 C \tag{29}$$

so the only Cauchy data living in C_{zz} not already present in $N_z z$ is the scalar $C(z,\bar{z})|_{\mathcal{I}^+_-}$. Finally, the Bondi mass $m_B(u,z,\bar{z})$ turns out to be almost completely constrained by the Einstein equations. In particular, the uu component of the Einstein equations can be expanding in powers of r, yielding at leading order[4]

$$\partial_u m_B = \frac{1}{4}\left[D_z^2 N^{zz} + D_{\bar{z}}^2 N^{\bar{z}\bar{z}}\right] - \tau_{uu}, \quad \tau_{uu} \equiv \frac{1}{4} N_{zz} N^{zz} + 4\pi G \lim_{r\to\infty}\left[r^2 T^M_{uu}\right]. \tag{30}$$

So given N_{zz}, its conjugate, and a known distribution of matter, m_B is determined up to an integration constant which we again take WLOG to be the values at $u = -\infty$. In short, we have

$$(\text{leading Cauchy data}) = \left\{N_{zz}(u,z,\bar{z}), C(z,\bar{z})|_{\mathcal{I}^+_-}, m_B(z,\bar{z})|_{\mathcal{I}^+_-}\right\}. \tag{31}$$

If we go to subleading order we get an infinite tower of further independent data, even after solving for the constraints implies by the Einstein equations. This is treated systematically by [**?**] and more recently by [**?**].

1.5 Scattering and Matching Conditions

All of the above analysis near $\mathcal{I}^+$ holds just as well near $\mathcal{I}^-$. Naively, it might seem that we could apply a BMS_+ transformation, and a simultaneous independent BMS_- transformation, and this would take us from one

[4] This is reasonably notation; the object is exactly the density outgoing flux of energy through $\mathcal{I}^+$ per solid angle and per retarded time.

symmetry to another (generally physically distinct) solution. But this cannot be right. For one, [**?**] notes that the gravitational scattering problem would be ill-posed, even if expressed in terms of purely small-gauge-invariant objects. More bluntly, in pure gravity the total outgoing flux of energy can be expressed as

$$E_{\text{out}} \propto \int_{\mathcal{I}^+} du d^2z \gamma_{z\bar{z}} N_{zz} N^{zz} \tag{32}$$

which changes under a generic supertranslation. If independent pairs of transformation in $\text{BMS}_+ \times \text{BMS}_-$ were symmetries, we could transform solutions that respect energy conservation to solutions that don't. It must be that applying a BMS_+ transformation is only a symmetry if we simultaneously apply a compensating BMS_- transformation, or equivalently that the physical symmetries are some diagonal subgroup of $\text{BMS}_+ \times \text{BMS}_-$; equivalently, it must be that choosing a BMS frame near one of $\mathcal{I}^+ \pm$ fixes the other.

In flat Minkowski space, we have the rule that $(z, \bar{z})$ in advanced and retarded coordinates refer to antipodally related points; in all popular treatments of gravity waves / perturbative gravity / weak-field gravity this is what is quietly done anyway by working in a fixed Minkowski background. [**?**] argues that this choice is equivalent to demanding

$$C(z,\bar{z})|_{\mathcal{I}^+_-} = C(z,\bar{z})|_{\mathcal{I}^-_+}, \quad m_B(z,\bar{z})|_{\mathcal{I}^+_-} = m_B(z,\bar{z})|_{\mathcal{I}^-_+} \tag{33}$$

and that this is likely the only Lorentz- and CPT-invariant choice.

2 $w_{1+\infty}$ and the Celestial Sphere

Scratch notes following [**?**].

2.1 Conformally Soft Gravitons

Denote by $G^+_\Delta(z,\bar{z})$ the celestial dual to a positive-helicity[5] bulk graviton. The authors of [**?**] find by analyzing bulk collinear limits that two such operators have OPE

$$G^+_{\Delta_1}(z_1,\bar{z}_1) G^+_{\Delta_2}(z_2,\bar{z}_2) = -\frac{\kappa}{2}\frac{1}{z_{12}} \sum_{n=0}^{\infty} B(\Delta_1 - 1 + n, \Delta_2 - 1) \frac{\bar{z}_{12}^{n+1}}{n!} \bar{\partial}^n G^+_{\Delta_1+\Delta_2}(z_2,\bar{z}_2) + \mathcal{O}\left(z_{12}^0\right). \tag{34}$$

Here B is the Beta function

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} \tag{35}$$

holomorphic poles[6]. These operators have conformal spin $s = +2$ and scaling weights Δ, hence conformal weights

$$\left(h, \bar{h}\right) = \left(\frac{\Delta + s}{2}, \frac{\Delta - s}{2}\right) = \left(\frac{\Delta}{2} + 1, \frac{\Delta}{2} - 1\right). \tag{36}$$

These operators are, by definition, the result of Mellin-transformating a bulk scattering amplitude of gravitons, necessarily in an asymptotically flat spacetime, to produce a boundary correlation function.

$$\langle G^\pm_\Delta(z,\bar{z}) \cdots \rangle \equiv \int_0^\infty \frac{d\omega}{\omega} \omega^\Delta \cdots \mathcal{A}\left(\omega \hat{p}(z,\bar{z}), \cdots\right). \tag{37}$$

From the well-known property of Mellin transformations

$$\int_0^\infty \frac{d\omega}{\omega} \omega^\Delta \left(\lim_{\omega\to 0} \omega f(w)\right) = \lim_{\Delta \to 1} (\Delta - 1) \int_0^\infty \frac{d\omega}{\omega} \omega^\Delta \omega f(w) \tag{38}$$

we see that a bulk amplitude involving a soft graviton

$$\lim_{\omega \to 0} \omega \mathcal{A}\left(\omega \hat{p}(z,\bar{z}), \cdots\right) \tag{39}$$

[5] TODO: Comment on choice to restrict to positive-helicity sector.

[6] TODO: Comment more on this choice.

corresponds on the boundary to taking the limit

$$\lim_{\Delta\to 1}(\Delta-1)G^+_\Delta(z,\bar z). \tag{40}$$

This nicely agrees with (34), which we see has an order-one pole at $\Delta_i = 1$, rendering this limit finite. In fact we see it has a tower of order-one poles at $\Delta_i = 1, 0, -1, \ldots$. We therefore define more generally

$$H^k(z,\bar z) \equiv \lim_{\varepsilon\to 0}\varepsilon G^+_{k+\varepsilon}(z,\bar z),\ \ k\in\mathbb{Z}. \tag{41}$$

We will shortly find that these live in a shortened representation, so that only a subset of k values give a nonzero result.

These objects turn out to generate an interesting algebra. We first expand in the antiholomorphic coordinate

$$H^k(z,\bar z) = \sum_n \frac{H^k_n(z)}{\bar z^{n+\frac{k-2}{2}}} \tag{42}$$

where the shift in the powers of $\bar z$ is required by H^k having a nonzero antiholomorphic conformal weight. Again, we will find that the relevant values of n are limited. We now interpret the objects $H^k_n(z)$ as holomorphic currents on the boundary. It will be illustrative to note the equivalent integral representation

$$H^k_n(z) = \oint_0 \frac{d\bar z}{2\pi i}\bar z^{n+\frac{k}{2}-2}H^k(z,\bar z). \tag{43}$$

2.2 Shortened Representations

To understand how these currents work, we compute their OPEs against celestial gravitons. Using the results of the last section, we find

$$\begin{aligned}
H^k_n(z_1)G^+_\Delta(z_2,\bar z_2) &= \oint_0 \frac{d\bar z_1}{2\pi i}\bar z_1^{n+\frac{k}{2}-2}H^k(z_1,\bar z_1)G^+_\Delta(z_2,\bar z_2)\\
&= \lim_{\varepsilon\to 0}\varepsilon\oint_0 \frac{d\bar z_1}{2\pi i}\bar z_1^{n+\frac{k}{2}-2}G^+_{k+\varepsilon}(z_1,\bar z_1)G^+_\Delta(z_2,\bar z_2)\\
&= \lim_{\varepsilon\to 0}\varepsilon\oint_0 \frac{d\bar z_1}{2\pi i}\bar z_1^{n+\frac{k}{2}-2}\left(-\frac{\kappa}{2}\frac{1}{z_{12}}\sum_{n=0}^\infty B(k+\epsilon-1+n,\Delta-1)\frac{\bar z_{12}^{n+1}}{n!}\bar\partial^n G^+_{k+\epsilon+\Delta}(z_2,\bar z_2)+\mathcal{O}\left(z_{12}^0\right)\right)\\
&= -\frac{\kappa}{2}\frac{1}{z_{12}}\lim_{\varepsilon\to 0}\varepsilon\sum_{n=0}^\infty\frac{1}{n!}B(k+\epsilon-1+n,\Delta-1)\bar\partial^n G^+_{k+\epsilon+\Delta}(z_2,\bar z_2)\oint_0\frac{d\bar z_1}{2\pi i}\bar z_1^{n+\frac{k}{2}-2}\bar z_{12}^{n+1}+\mathcal{O}\left(z_{12}^0\right).
\end{aligned} \tag{44}$$

The innermost integral can be evaluated by binomial expansion as

$$\begin{aligned}
\oint_0\frac{d\bar z_1}{2\pi i}\bar z_1^{n+\frac{k}{2}-2}(\bar z_1-\bar z_2)^{n+1} &= \sum_{\ell=0}^{n+1}\binom{n+1}{\ell}(-\bar z_2)^{n+1-\ell}\underbrace{\oint_0\frac{d\bar z_1}{2\pi i}\bar z_1^{n+\frac{k}{2}-2}\bar z_1^\ell}_{=\delta_{n+\ell+\frac{k}{2}-2,0}}\\
&= \sum_{\ell=0}^{n+1}\frac{(n+1)!}{\ell!(n+1-\ell)!}(-\bar z_2)^{n+1-\ell}\delta_{n+\ell+\frac{k}{2},2}.
\end{aligned} \tag{45}$$

For k, n such that this is ever nonzero, the only contributing term in the sum is

$$\ell = 2-n-\frac{k}{2}\implies\binom{n+1}{\ell} = \frac{(n+1)!}{\left(2-n-\frac{k}{2}\right)!\left(2n+\frac{k}{2}-1\right)} \tag{46}$$